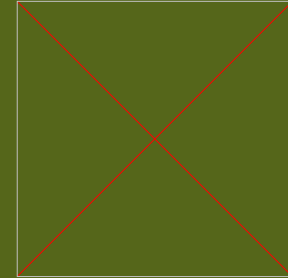




INTEGRATING HPC, AI & SCIENCE GATEWAYS INTO THE CAP 4620 ARTIFICIAL INTELLIGENCE COURSE



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ABSTRACT

Through FacultyHack@Gateways 2026, I redesigned CAP 4620: Artificial Intelligence at Florida A&M University to bring shared research cyberinfrastructure into an undergraduate AI course that had previously relied on local machines and Google Colab. The redesign integrates the National Research Platform (NRP) — a national Kubernetes-based platform of shared CPU/GPU resources — as the primary computing environment, with the NRP JupyterHub serving as the students’ science gateway: they select CPU/GPU and memory resources, launch GPU-accelerated Jupyter notebooks, execute ML workflows, monitor jobs, and retrieve and visualize results without touching the underlying infrastructure. Rather than a standalone HPC unit, cyberinfrastructure is embedded in existing topics using the course’s TensorFlow and scikit-learn stack: students run the same neural-network experiment on CPU versus GPU, train CNN image classifiers on NRP GPUs while varying batch size, epochs, and architecture, launch hyperparameter configurations as parallel experiments, and complete a structured NRP-versus-Colab comparison documenting when shared research cyberinfrastructure outperforms free cloud notebooks. The redesign also expands “optimization” to include computational performance — training and inference time, CPU/GPU utilization, memory, and scalability — and rebuilds the final project (beginning after Week 8) as an AI + HPC experience: students execute experiments on NRP, analyze both model and computational performance, and publish a reproducible GitHub workflow. Explicit HPC and science-gateway learning outcomes are mapped to labs and the project, making cyberinfrastructure assessable rather than supplemental, with Cornell Virtual Workshop and HPC Carpentry scaffolding HPC concepts. This mattered because most FAMU computing undergraduates had never touched research cyberinfrastructure; the gateway let them work in a familiar notebook environment while gaining authentic experience with resource selection and performance — skills that connect to graduate research and industry AI work and strengthen the pipeline our AI & Data Science Cluster is building. Two honest lessons emerged. First, less is more: focusing on a single environment (NRP) rather than surveying several platforms gave students a deeper, more confident experience — the temptation to showcase everything is real, and resisting it made the course better. Second, cyberinfrastructure onboarding takes longer than expected; account setup, resource selection, and environment configuration deserve dedicated lab time early in the semester, not a quick walkthrough squeezed into an existing lesson.

INTRODUCTION

CAP 4620 is FAMU’s upper-level AI course, a key elective in the department’s AI and data science pathway. Before this redesign all ML work ran on laptops or Google Colab, and most students had never touched research cyberinfrastructure — skills expected in graduate research and industry AI. This project integrates the NRP Jupyter Hub as the science gateway providing students access to GPU-accelerated research cyberinfrastructure. NRP lowers the barrier to entry so students can engage with real LLMs almost immediately, making it a strong fit for AI-literacy courses like Artificial Intelligence CAP 4620.

Course Goals

- Master core AI/ML concepts — search, neural networks, deep learning, model evaluation — in Python
- Analyze how batch size, architecture, and CPU vs. GPU choices affect training time and performance
- Practice reproducibility: Git, documented environments, published workflows
- Execute GPU-accelerated AI/ML workflows on the NRP JupyterHub

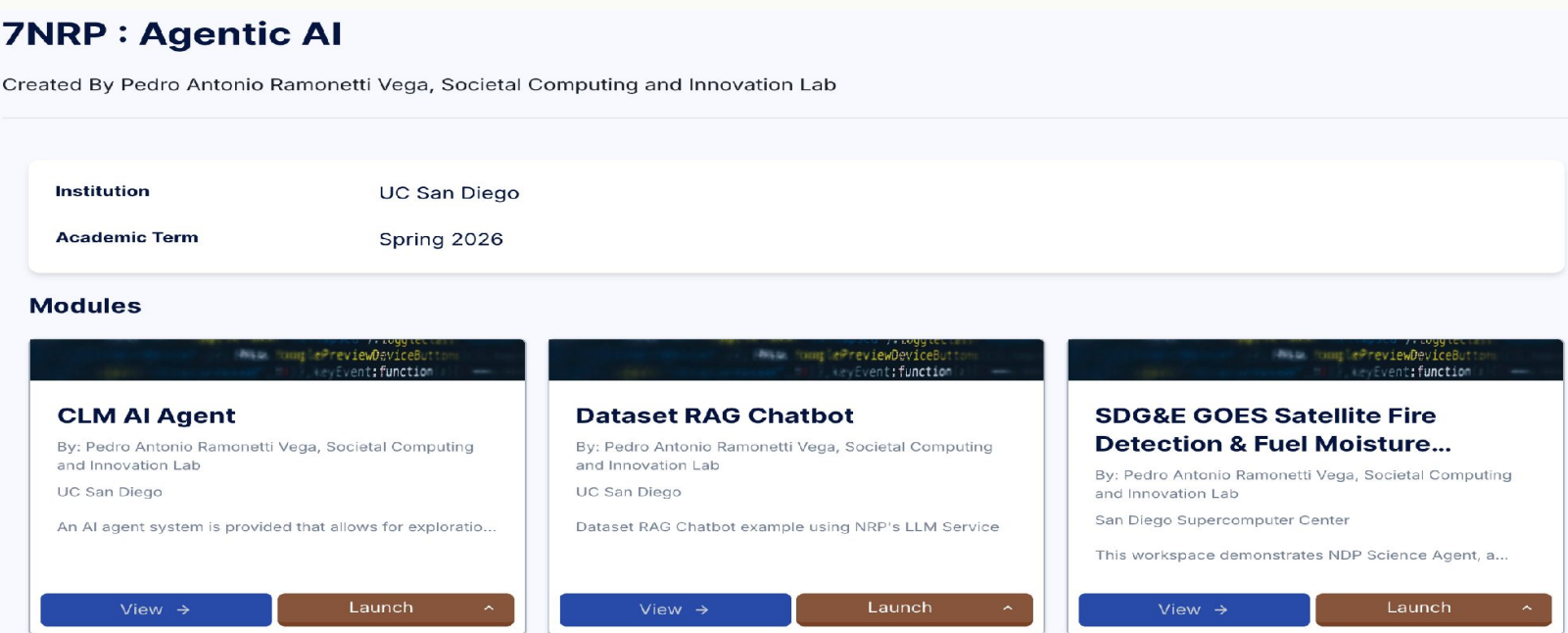
Grading Scale

A	90% – 100%
B	80% – 89%
C	70% – 79%
D	60% – 69%
F	59% and Below

Mentor Feedback Incorporated

- Embed HPC into existing AI/ML topics rather than a standalone unit — led to CPU vs. GPU, CNN training, and parallel hyperparameter labs built into current course topics
- Give students a science gateway experience — led to adopting the NRP JupyterHub as the interface for selecting resources and executing workflows
- Make cyberinfrastructure assessable — led to explicit HPC/gateway learning outcomes, computational performance analysis, and an AI + HPC final project

Datasets

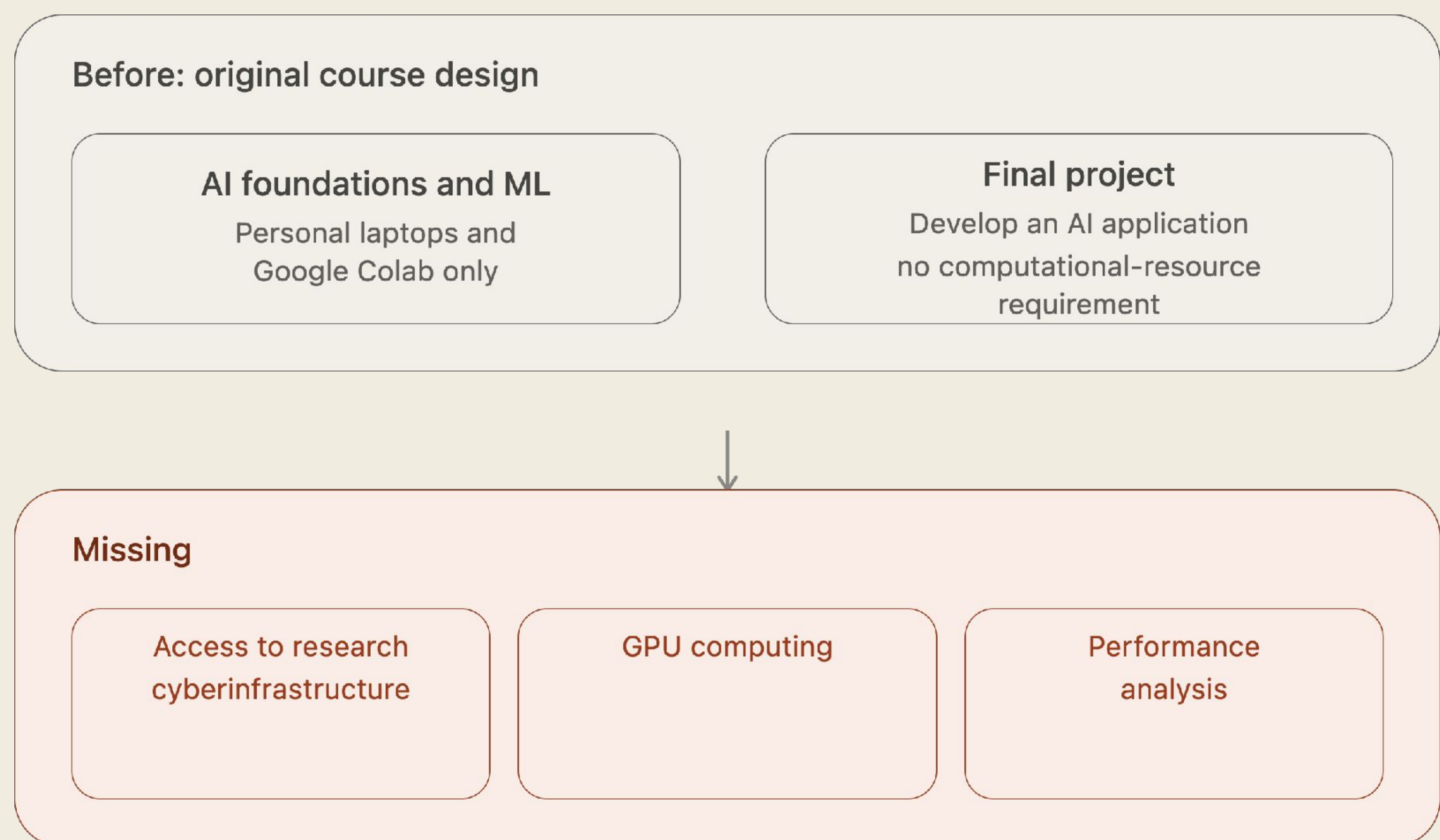


IMPACT ON STUDENTS

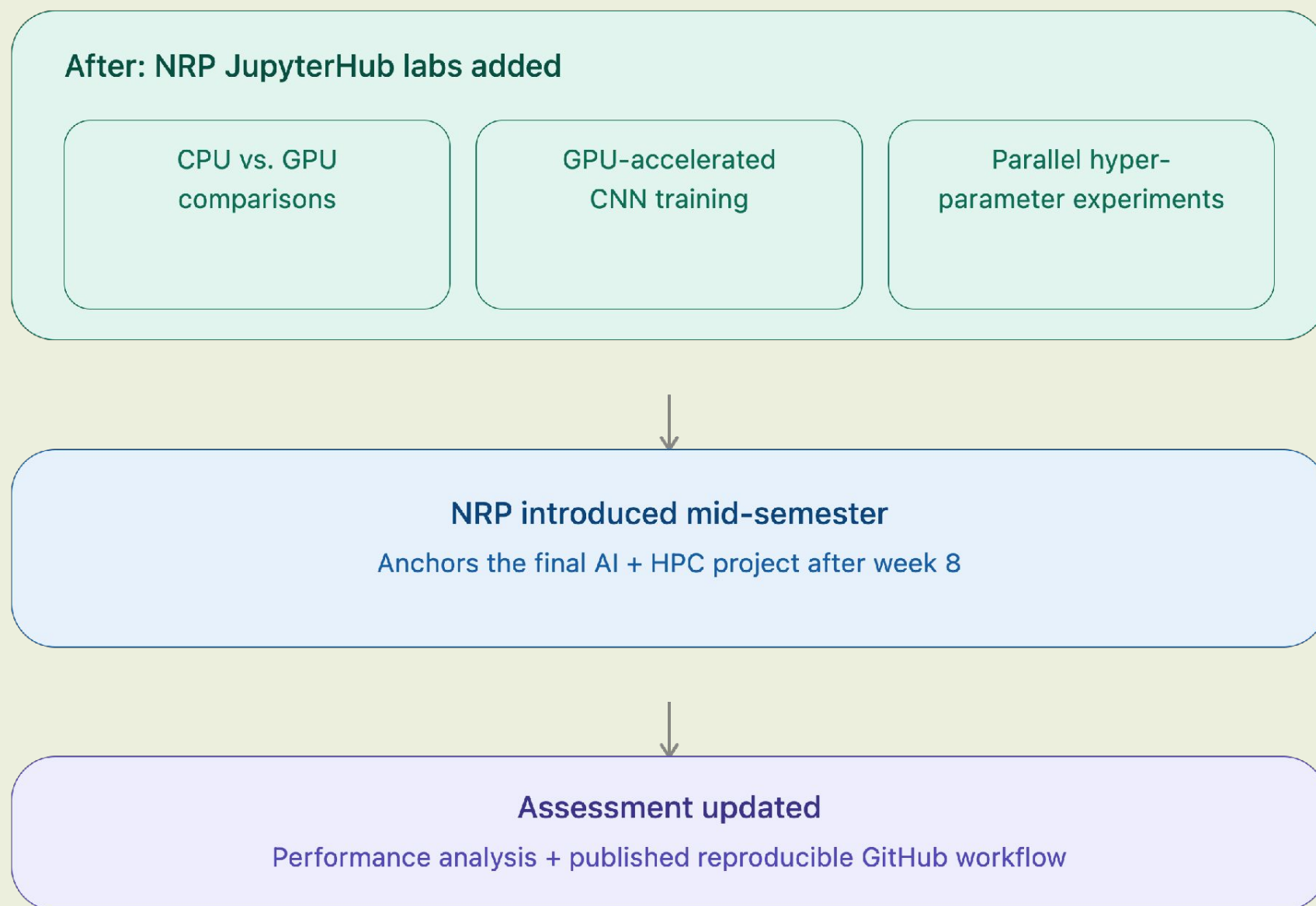
Users are undergraduate computing majors in CAP 4620 at FAMU, most of whom had never used research cyberinfrastructure before this course. They gained GPU computing experience, confidence selecting and monitoring computational resources, and a published, reproducible AI workflow on GitHub — access and skills that connect directly to graduate research and industry AI roles and strengthen the department’s AI and data science pipeline.

COURSE REDESIGN: BEFORE & AFTER

INITIAL SYLLABUS



MODIFIED SYLLABUS



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PROJECT GITHUB REPO

github.com/Yohn-Parra

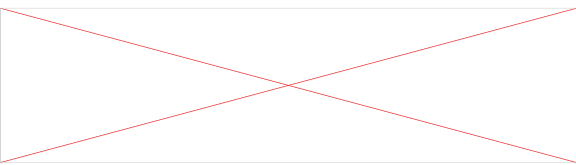
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